AJEENKYA
D Y PATIL UNIVERSITY
THE INNOVATION UNIVERSITY

A
MINI PROJECT REPORT ON
Predicting Diabetes Using Machine Learning

FOR

Term Work Examination

Bachelor of Computer Application in Artificial Intelligence and Machine learning (BCA - AIML) Year 2024-2025

Ajeenkya DY Patil University, Pune

-Submitted By-

Mr.

Under the guidance of

Prof. Vivek More

Ajeenkya DY Patil University

D Y Patil Knowledge City,
Charholi Bk. Via Lohegaon,
Pune - 412105
Maharashtra (India)

.

Date: 14/04 / 2025

CERTIFICATE

This is to certified that Ketan Harde.
A student's of **BCA(AIML) Sem-IV** URN No 2023-B-22112003
has Successfully Completed the Dashboard Report On

Predicting Diabetes Using Machine Learning

As per the requirement of
Ajeenkya DY Patil University, Pune was carried out under my supervision.
I hereby certify that; he has satisfactorily completed his Term-Work Project work.

Place: - Pune

Examiner

INDEX	
	Page No.

Mini Project Report

Project Title :-

Predicting Diabetes Using Machine Learning.

Introduction:-

Diabetes is one of the most prevalent chronic diseases up to today as it affects millions of people worldwide. When the body fails to produce or efficiently utilize insulin, blood sugar levels become abnormal. Early diagnosis of diabetes may keep patients from experiencing adverse effects, such as kidney failure, blindness, and heart disease.

To construct a machine learning (ML) model that establishes the prediction of whether a person has diabetes based on certain parameters of diagnosis, the project. The database which is referred to collectively for the purpose of this project is the Pima Indians Diabetes Dataset; it consists of records of women only above 21 years in age with Pima Indian heritage. The dataset includes parameters such as glucose level, blood pressure, BMI, insulin level, etc.

Machine learning often combined with effective data preprocessing and visualizations becomes formidable tools to trace patterns in medical data showing model creation which can output accurate forthcoming health predictions.

Diabetes is perhaps the most widely known chronic disease today, affecting millions of people around the world. Insufficient or ineffective insulin production in the body causes abnormal blood sugar levels. Early detection of diabetes can prompt timely preventive action by the patient, especially to prevent possible serious complications like kidney failure, blindness, or heart disease.

The project will thus build a machine learning (ML) model that predicts whether an individual has diabetes based on some diagnostic measures. For this endeavor, the data being taken is the Pima Indians Diabetes Data set, which is data concerning Pima Indian women that are 21 years and older. This data set includes parameters such as glucose levels, blood pressure, BMI, insulin level, etc.

Objective:-

The specific objective of the present project shall be the realization of the machine-learning model capable of predicting the diabetes presence in an individual by diagnostic medical data. Such objectives shall apply to this project:

1. Understanding and preprocessing the dataset, including handling missing or invalid values, normalizing features, and preparing the data for analysis and modeling.

2. Perform exploratory data analysis (EDA) to identify patterns and relationships between features with visualizations and statistical methods, as well as any correlation which may exist between features.

3. Training and evaluating a machine-learning model, Logistic Regression, to classify persons into either diabetic or non-diabetic.

4. Visualization of results from interpreting model performance and feature importance using graphs such as heat maps, confusion matrixes, and accuracy plot.

5. The potential insights will be supplied to assist health practitioners to be on the lookout for these diabetes risks, opening avenues for future research or real-time prediction applications.

Methodology:-

Every method applied in this process, covering from data collection to model evaluation are mentioned in this section.

1. **Data Collection**

The dataset used is a public one: The Pima Indians Diabetes Dataset, which has 768 records having 8 input features and 1 outcome class (Outcome: 0 for non-diabetic; 1 for diabetic).

2. **Data Preprocessing**

We do the following on data before being entered into the machine learning model:

- To Handle Missing Values: Replacing 0 in some of the feature values (say glucose and insulin) with NaN is a practice, followed by imputing it with the feature median.

- Normalization: Scaling features to bring all features into a comparable scale using StandardScaler.

3. **Exploratory Data Analysis (EDA)**

EDA has a role in understanding the patterns and relationships existing in the data, such as:

- Data distribution check.
- Visualization of the correlation between attributes using heat maps.
- Histogram and boxplot to study outliers.

4. **Model Selection**

We carry out Logistic Regression; a straightforward and effective classification algorithm well suited to binary classification tasks. The probabilities of an outcome are therefore modeled using a logistic function.

5. **Model Evaluation**

The evaluation model process included the separation of data into training and test groups, and tried to assess model performance through:

- Accuracy Score.
- Confusion Matrix.
- Precision, Recall, F1 Score.

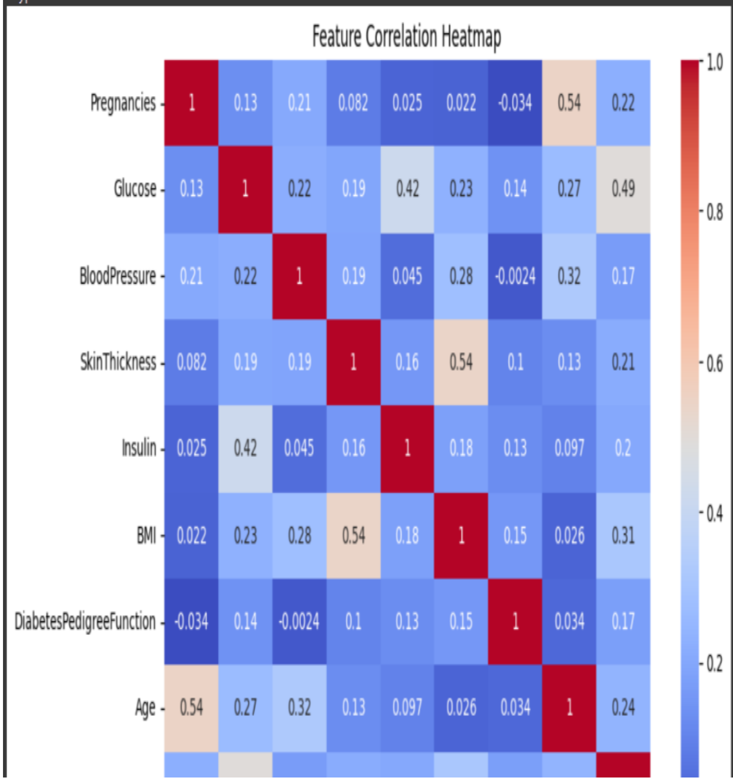
Implementation And Code:-

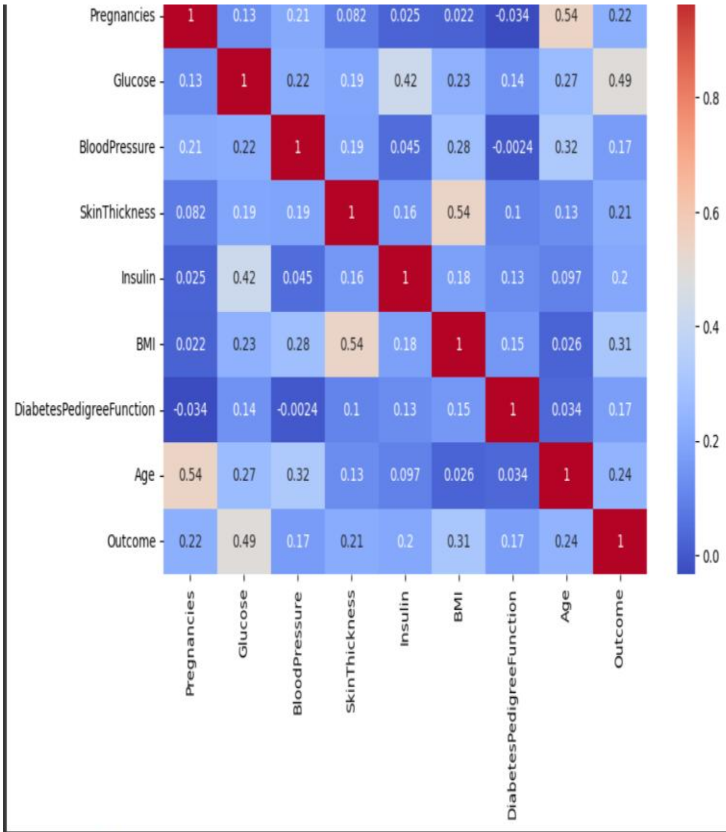
```
# Importing necessary libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import classification_report, confusion_matrix, accuracy_score

# Load dataset
url = "/content/diabetes.csv"
df = pd.read_csv(url)

# Display first few rows
print(df.head())

# Check for missing values
print(df.isnull().sum())

# Replace 0 values in some columns with NaN
cols_with_zeros = ['Glucose', 'BloodPressure', 'SkinThickness', 'Insulin', 'BMI']
df[cols_with_zeros] = df[cols_with_zeros].replace(0, np.nan)

# Fill missing values with median
df.fillna(df.median(), inplace=True)

# Correlation heatmap
```

```python
df.fillna(df.median(), inplace=True)

# Correlation heatmap
plt.figure(figsize=(10,6))
sns.heatmap(df.corr(), annot=True, cmap='coolwarm')
plt.title("Feature Correlation Heatmap")
plt.show()

# Data splitting
X = df.drop("Outcome", axis=1)
y = df["Outcome"]

# Feature scaling
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

# Train-test split
X_train, X_test, y_train, y_test = train_test_split(X_scaled, y, test_size=0.2, random_state=42)

# Model training
model = LogisticRegression()
model.fit(X_train, y_train)

# Predictions and evaluation
y_pred = model.predict(X_test)
print("Accuracy:", accuracy_score(y_test, y_pred))
print(confusion_matrix(y_test, y_pred))
print(classification_report(y_test, y_pred))
```

Results And Visualisation:-

```
   Pregnancies  Glucose  BloodPressure  SkinThickness  Insulin   BMI  \
0            6      148             72             35        0  33.6
1            1       85             66             29        0  26.6
2            8      183             64              0        0  23.3
3            1       89             66             23       94  28.1
4            0      137             40             35      168  43.1

   DiabetesPedigreeFunction  Age  Outcome
0                     0.627   50        1
1                     0.351   31        0
2                     0.672   32        1
3                     0.167   21        0
4                     2.288   33        1
Pregnancies                 0
Glucose                     0
BloodPressure               0
SkinThickness               0
Insulin                     0
BMI                         0
DiabetesPedigreeFunction    0
Age                         0
Outcome                     0
dtype: int64
```






```
Accuracy: 0.7532467532467533
[[82 17]
 [21 34]]
              precision    recall  f1-score   support

           0       0.80      0.83      0.81        99
           1       0.67      0.62      0.64        55

    accuracy                           0.75       154
   macro avg       0.73      0.72      0.73       154
weighted avg       0.75      0.75      0.75       154
```

Conclusion & Future Scope:-

📃 Conclusion:-

This project basically shows a typical machine learning pipeline applied to build a predictive model solving a real-world medical problem. With appropriate data preprocessing, visualization, and model selection, we could get some promising results. Logistic Regression has been found to be suitable for this classification task, especially considering its simplicity and its interpretability.

🚀 Future Scope:-

- Advanced Model: Use Random Forest, SVM, or Neural Network for the purpose of increasing accuracy.
- Hyperparameter Tuning: Use GridSearchCV to find the best parameters.
- Feature Engineering: This includes new features such as age group, BMI categories, etc.
- Deployment: Web application deployment using Streamlit/Flask.
- Real-time prediction: Connect with real-time health monitoring apps or devices.